Assignment-02

Unit-01

Page No.
Date 24 09 2024

2m

① Define Internet and briefly explain its role in WWW.

⇒ Internet is an abbreviation for Interconnected NETwork, is a wide area network that connects millions of computing devices across the world. It is a global network of networks that facilitates data transfer & communication services ~~via web~~. Carries vast array of information resources & services, the inter-linked hypertext documents of WWW & the infrastructure to support Email.

② What is the function of Web browser in the context of WWW.

⇒ Web browser is used to access information on WWW, as that extracts information on user request from the internet and presents it as a webpage to the user.

③ Differentiate b/w Web browser and Web server.

⇒

Web Browser	Web Server
* Web browser acts as an interface b/w the server and the client and displays a web document to the client	* Web server is a software or a system which maintain the web applications, generate response and accepts clients data.
* Browser runs on client-side	* ~~Browser~~ Server runs on server side.

④ Explain the purpose of URL in Web navigation.

⇒ Uniform Resource Locator is the address of a website, used to access resources on the web, such as web pages, images or a file on the Internet...

Syntax:- scheme://hostname:port/path-file-name

→ It serves several key purpose in web navigation.

① Identifying Resources → Each URL uniquely identifies a specific resources on the internet.

② Locating Resources → Specifies Domain name or IP address of the server that hosts the resource, as well as the specific path to the resources.

③ Enabling Web Requests → When you enter URL into browser, it sends request to the server where the resource is stored.

④ Navigating b/w Webpages → URL's allow users to easily navigate b/w different pages on the web, through hyperlinks.

Page No.
Date

(5) Define MIME, & its significance in web communications.

⇒ MIME (Multipurpose Internet Mail Extensions) is a standard that was originally designed for email to allow the transmission of various types of data (text, image, audio, video) over the internet. It's use has expanded beyond email and is now integral to web communication as well.

→ Significance of MIME in Web Communications:-

ⓐ HTTP and Web browsers.

→ When a web server sends a file to a web browser, it includes a MIME type in HTTP response headers to tell the browser what kind of file it is receiving and how to handle it.

For ex • Content-Type: text/html → tells the browser to render the response as an HTML page.
• Content-Type: application/pdf → tells the browser to display or download the files as a PDF document.

ⓑ File uploads

→ When user upload files through web forms, the MIME type is sent to the server to identify the format of the uploaded files. This helps the server process the file correctly.

ⓒ Email Attachments

→ In email communications, MIME allows users to send attachments alongside plain text content, videos, images,

ⓓ API's and Web services

→ When interacting with web services and API's, MIME types are used to specify the format of request and response data.

→ application/json ⇒ is used for JSON-based API's

(6) Explain Internet security (IOS). Explain with an example.

⇒ Internet security refers to the measures taken to protect computer systems and networks from unauthorized access, theft, damage or disruption of services. Internet security involves a range of technologies, policies that are designed to prevent, detect, and respond to these threats.

Ex: Internet security measures include firewalls, antivirus software, encryption, multifactor authentication, regular software updates and employee training. By implementing these measures, organizations & individuals can reduce the risk of cyber attacks and protect their online assets.

Scanned with OKEN Scanner

Page No.
Date

⑦ What do you means by web Programming tool box.

⇒ Collection of diverse tools that are essential for creating dynamic and engaging online experience. It includes foundational markup languages like HTML and XML, versatile programming language like JS, server-side languages like PHP and Ruby, frameworks like JSF, ASP.NET, and Rails, multimedia technologies like Flash, Ajax for asynchronous data exchange..

This tool box equips web programmers with the necessary tools to navigate the complexities of modern web development.

⑧ Briefly describe JS execution environment.

⇒ JS execution environment refers to the environment in which JS code is executed. It is typically the client's browser. The window object and document object are two important objects in JS execution environment.

⑨ Explain DOM.

⇒ DOM is a way of representing the document independent of browser type. It allows a developer to access the document via a common set of objects, properties, methods, and events & to alter the contents of the web page.

⑩ Name one object commonly accessed using the navigator object in JS.

⇒ navigator.javaEnabled( ).

5M

① Discuss the fundamentals of WWW, highlighting the roles of the internet, web browsers & web servers.

⇒ World wide web is a system of interlinked hypertext documents accessed via the internet.

⇒ It allows users to view and navigate information through web browsers. The web operates using several technologies and follow certain rules and protocols to ensure seamless communication between browsers and servers.

* Internet rules and protocols

→ web relies on various rules and standards, while enable browsers and servers to communicate.

Scanned with OKEN Scanner

→ HTTP/HTTPS

- HTTP is the foundational protocol used to transmit data over the web. It allows web browsers to request web pages from servers.
- HTTPS is the secure version of HTTP, using SSL/TLS encryption to protect data in transit.

→ URL

- Uniform Resource Locator is a web address used to locate a resources on the web.
- It consists protocol: // domain name / path / filename.

→ HTML

- Standard Markup Language used to create web pages.
- HTML is used to create web pages.

→ DNS

- Domain Name System translates human readable domain names into IP address, that computers use to locate servers on the internet.

* Web browsers

- Software Application that allows users to access, retrieve and view web pages on the internet.
- Browsers interpret and render HTML, CSS, JS & other web technologies.
- Working :-
  - Sends requests to web servers
  - Render web content
  - Support navigation
- Ex:- Google Chrome, Mozilla FireFox, Microsoft Edge, Safari.

* Web Servers

- A system, that hosts websites and serves web content to clients.
- Working :-
  - Receive requests from browsers.
  - Serve files and applications
  - Ensure security and manage resources.
  - Ex:- Apache HTTP Server, NginX.

The interaction between clients and servers over the Internet is what makes world wide web functional.

Scanned with OKEN Scanner

Page No.
Date

② Explain the concept of URL & How they contribute to efficient web navigations.

⇒ URL (Uniform Resource Locator)

- URL is a web address used to locate a resources on the web.
- Syntax of URL

Scheme://hostname:Port/path-and-Filename

① Scheme - Tells the browser how to deal with the file that it is about to open. The most common scheme, we will see is http, https, FTP, mailto etc.

② Hostname - DNS domain name of the server.

③ Port - TCP port number, that the server is listening for incoming requests from the clients.

④ Path-and-Filename - Location of the requested resource under the server document base directory.

Ex:- http://www.w3schools.com/HTML tutorial.

→ It serves several key purpose in web navigation.

① Identifying Resources → Each URL uniquely identifies a specific resources on the internet.

② Locating Resources → Specifies Domain Name or IP Address of the server that hosts the resources, as well as the specific path to the resources.

③ Enabling Web Requests → When you enter URL into browser, ~~it sends requ~~. user can navigate to specific resources without needing to rely on links.

④ Hyperlinking → URLs are embedded within websites as hyperlinks, enabling easy navigation b/w pages or websites.

⑤ Reference → URLs are used to share and bookmark specific web content, ensuring easy access later.

Essentially, URLs are the foundation of web navigation acting as the "address" of content and enabling seamless access to resources online.

Scanned with OKEN Scanner

(3) Describe the role of MIME in ensuring proper communication b/w different Web entities.

Page No.
Date

=> MIME (Multipurpose Internet Mail Extension) plays a critical role in ensuring proper communication b/w different web entities by defining a standardized way to encode and structure various types of data.

-> Although originally developed for email, MIME is now widely used in web communications, such as HTTP, to support the transmission of various media types.

-> Here's how MIME ensures effective communication:-

- Content-Type Identification:-

* MIME assigns Content-Type header that tells the receiving entity what type of data is being transmitted. This could be text, images, audio, video or application data. (JSON / XML).

for example, Content-Type : image/png indicates a PNG image. Content-Type : text/html indicates that the content is HTML. This allows web servers and browsers to understand how to interpret and display the content properly.

- Data Encoding:-

* MIME defines encoding schemes like Base64 to convert binary data (img/file) into text format, allowing binary data to be transmitted over text-based protocols like HTTP / SMTP without data corruption. For an instance, an image can be encoded as Base64 and sent as text.

- List of some commonly used MIME Types

MIME TYPE	Type of file	Extensions
text/html	Hyper text markup lang	html, .html
text/css	webpage might link to an external stylesheet for styling	.css
application/javascript	web page includes interactive features through JS	.js
image/jpeg & /png	Image based on their formats	.jpeg, .png
audio/mp3 & video/mp4	Audio Video	.mp3 .mp4
application/pdf	Adobe Portable Document Format	.pdf

Scanned with OKEN Scanner

Overall, MIME enables diverse content to be reliably transmitted across the web facilitating proper understanding and rendering of different media by web browsers, servers and email clients.

4) Explore the importance of Internet Security in the context of web.

⇒ • Protection of Personal Information:-

- Internet security is crucial for safe guarding personal and sensitive information such as credit card details passwords and personal identification.
- Unauthorized access to this data can lead to identity theft, financial loss and privacy breaches

• Safeguarding Intellectual property Rights:-

- Internet security is essential for protecting intellectual property, trade secrets and proprietary information from unauthorized access or theft.

• Compliance with Regulations:-

- IS is essential for ensuring compliance with data protection and privacy regulations.
- Legal requirements dictate how organizations handle and protect user data, & failure to comply can result in legal consequences and penalties.

• B/S continuity:-

- Internet security is vital for ensuring the uninterrupted operation of online services, e-commerce platforms and digital business processes.
- Any disruption in services due to security breaches or cyber attacks can result in financial losses and damage to the reputation of b/s.

• Trust and Confidence:-

- Internet security plays a key role in building and maintaining trust among users.
- Individuals are more likely to engage with online platforms, share information, and conduct transactions if they have confidence in the security measures implemented by service providers.

Scanned with OKEN Scanner

⑤ Briefly explain Web programmer tool box with ex.

Page No.
Date

⇒ • HTML :-

→ Markup language used for creating web pages.
→ It provides a set of instructions for web browsers on how to display documents, rather than performing computations.
→ Term "markup" originates from the publishing domain, where it signifies the instructions given to a printer for rendering text, graphics & other elements in a specific manner.
→ HTML uses tags to define the structure and layout of web content.
→ Tag wraps around specific content and acts as commands for browsers.
→ HTML guides web browser to display web content correctly.
→ Tools used are:- HTML Editors, WYSIWYG Editors.

• Plug-ins and Filters :-

→ Plug-ins are programs that can be integrated with a word processor to add new capabilities to the word processors.
→ Plug-ins provide convenient ways to insert HTML into document being created or edited. This makes the word processor appear to be an HTML editor that provides WYSIWYG (What You SEE Is What You Get) HTML document development.
→ End result of using a plug-in to create HTML docs is an HTML file that can be viewed in a web browser.

• XML

→ eXtensible Markup Language is a simplified version of SGML (Standard Generalized Markup language).
→ Designed to allow users to create their own custom markup languages.
→ XML allows users to create tags and attributes that are specified to their needs.
→ XML documents can undergo validation to ensure that they conform to the standard form for their respective markup languages, simplifying the development of application programs.

Scanned with OKEN Scanner

• JS :-

→ JS is a programming language that is often used for client-side scripting in web programming.
→ It is typically embedded in HTML documents and is interpreted by the client's browser.
→ JS can be used for a variety of purposes, such as validating form data, creating dynamic HTML docs, and building Ajax-enabled HTML.
→ JS is often used in conjuction with HTML and CSS to create interactive and engaging web pages.

• PHP :-

→ Open source scripting language well suited for web development and can be embedded into HTML.
→ PHP code is executed on the server, generating HTML which is then sent to the client.
→ This means the client receives only the output of the PHP code and not the actual code itself.
→ PHP offers ease of use and flexibility along with built-in modules for accessing user input, data, manipulating arrays and strings, interacting with various databases and much more.
→ functionality like database connections, file manipulation, session tracking

• AJAX (Asynchronous JavaScript and XML)

→ Creates asynchronous web applications.
→ With AJAX, small bits of data are exchanged with the server behind the scenes, without reloading the entire web page. This enables rich, dynamic interfaces and seamless user experiences in the modern web Applications.
→ AJAX uses JS to make background HTTP requests to the web server and process the received data before selectively updating parts of the webpage.

• ASP.NET :-

→ Active Server Pages .NET is a server-side web application framework designed for building dynamic web sites, web applications & web services.
→ ASP.NET is built on .NET framework and is the successor to MS original Active server pages (ASP) technology.

Scanned with OKEN Scanner

⑥ Explain JavaScript execution environment and it's types of accessibility.

Page No.
Date

→ JavaScript execution environment refers to the environment in which JS code is executed.
In web programming, JS execution environment is typically the client's browser. The window object & document object are 2 objects in JS execution environment.

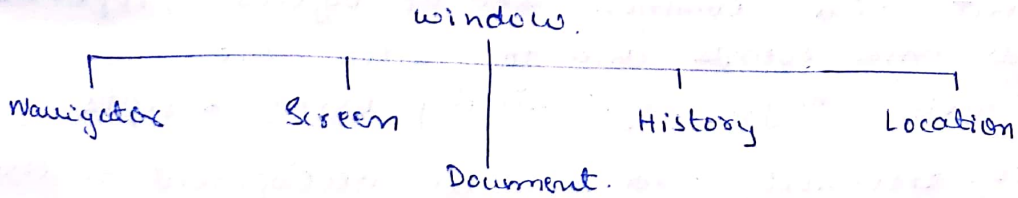
* window object
* Document object.

* window object

→ The browser provides global object called the window object.
→ Represents the browser window and provides access to various properties and methods.
→ The window object is the top-level object in the JavaScript hierarchy and is used to interact with the browser and user's computer.
→ The window object are visible to all JS script that appear in HTML document. When a global variable is defined in JS code, it is implicitly created as a new property of the window object.

Property	Method
window.innerHeight	window.alert(message)
window.screenX	window.confirm(message)
window.outerWidth	window.prompt(msg, default)
window.location.href	window.location.replace(url)
window.navigator.userAgent	window.open(url, target, features)

Basic structure of the object model in modern web browser

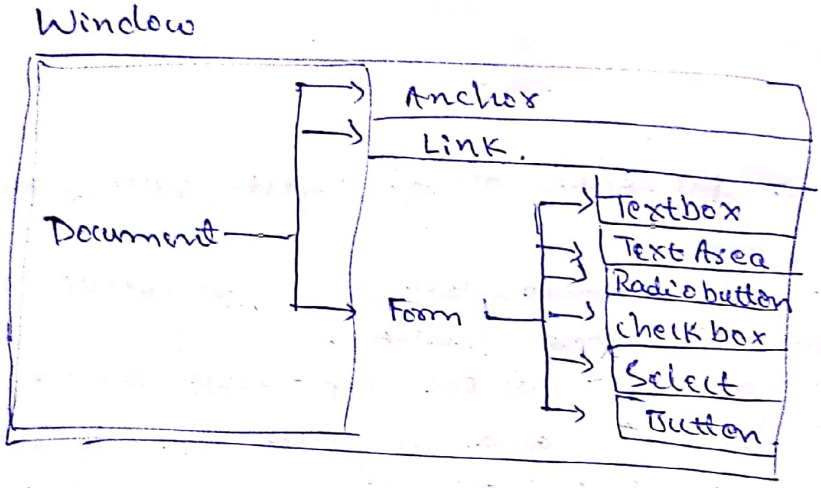


Scanned with OKEN Scanner

*Document object

-> Represents the web page that is currently loaded in the browser window and provides access to the various elements on the page, such as forms, buttons and text boxes.

-> Document object is a property of the window object and can be accessed using window.document property.



Property	Method
document.URL	document.getElementById(id)
document.forms	document.appendChild(node)
document.cookie	document.getElementsByClassName(tag)
document.links	document.getElementsByTagName
document.stylesheets.	document.removeChild(node)

④ Explain DOM and DOM2

⇒ DOM is a way of representing the document independent of browser type. It allows a developer to access the document via a common set of objects, properties, methods and events and to alter the contents of the web page dynamically using scripts.

* DOM is extensively used in web development to create dynamic and interactive web pages as :-
- Changing the content of a webpage dynamically using JS on user's input.
- Adding or removing elements from a web page based
- Responding to user events like clicks and key presses.
- Validating form input before submitting it to server

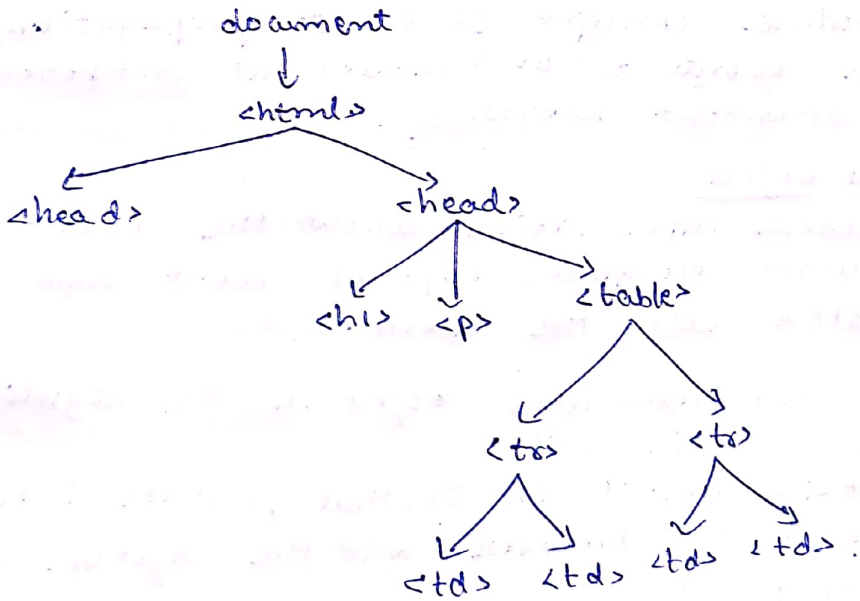
- Animating elements on a webpage using CSS and JS.

Page No.
Date

→ Levels of DOM

- **DOM0**
  → JS has 1st introduced & web browsers had limited support for the DOM.
- **DOM 1**
  → Recommended by W3C in late 1988.
  → It includes basic features like the ability to access and manipulate elements on a web page.
- **DOM 2**
  → This level adds more advanced features like support for events, style sheets and XML namespaces, HTML forms & tables.
- **DOM 3**
  → It is intended to resolve a lot of the complications that still exist in the event model in level2 of the standard and adds support for XML features.
  → This level adds even more advanced features like Support for xpath, keyboard navigation and validation.

→ Tree representation of DOM



Scanned with OKEN Scanner

⇒ DOM 2

- DOM 2 events model is a standard for handling events in web browsers.
- DOM 2 event model defines a standard set of events types and provides a way to register event listener for those events.

DOM 2 introduces the concept of event propagation, which allows events to be handled at different levels of the document hierarchy. It is mainly used in web development.

→ Key features of the DOM 2 Events Model

① Standard Event Types.

- Defines standard set of event types such as click, mouseover, and key down. This makes it easier for developers to write cross-browser code that works consistently across different platforms.

② Event listeners

- DOM 2 event model provides a way to register event listeners for specific events. This allows developers to write code that responds to user actions, such as clicking a button or typing in a text field.

③ Event Propagation

- Introduces concept of event propagation, which allows events to be handled at different levels of the document hierarchy.

④ Event object

- Contains information about the event, such as the target element, type of event and any data associated with the event.

⑤ Explore the navigator object in JS, highlighting it's practical uses in WP

⇒ • A Built-in object in JS, that provides information about the web browser and the system on which it is running.

- Main purpose of navigator object is to get information about the browser, so that web developers can make decisions based on client's environment. This can include adjusting the website's behaviour for compatibility, offering downloads suitable for the user's platform.

Scanned with OKEN Scanner

→ Methods

Page No.
Date

navigator.appName → return Netscape.
navigator.appVersion → Indicates browser version & platform.
navigator.language → Preferred language of the user
navigator.geolocation → Geographical object to locate user's position
navigator.cookieEnabled → return 'true' if cookies are enabled
navigator.javaEnabled() → return 'false' if Java is not supported.

→
```
<!DOCTYPE html>
<html>
<head>
    <meta charset = "UTF-8">
    <title> JS Navigator object </title>
    <script type = "text/ JavaScript">
        function showProperties()
        {
            alert ("Name of the browser is :" +navigator.appName
                + "\n" + "The version of the browser is :" +navigator.appVersion
                + "\n" + "online :" + navigator.online+
                "\n" + "The cookies enabled :" +navigator.cookieEnabled + "\n" + "The platform of the client is :" + navigator.platform);
        }
    </script>
</head>
<body>
    <form id = "myform">
        <input type = "button" value = "click me"
        onclick = "showProperties()"> </form>
</body>
</html>
```

→ Output

CLICK ME

Name of the browser is : Netscape.
The version of the browser is : 5.0 (windows).
online : true
The cookies enabled : true.
The platform of the client is : Win32

OK

SM.

Page No.
Date

1). Develope a comprehensive guide on Internet Security practices For web developers.

=> Web developers ~~defer~~ are the first line of defence when building secure applications. Neglecting security can lead to serious breaches, data theft and reputational damage.

Guidelines on Internet Security are:

- Use HTTPS.
  -> Secure protocol, encrypts data b/w the client and server, preventing interception.
  -> Get an SSL/TLS certificate. Enforce HTTPS by using HTTP strict Transport Security (HSTS).
- Input Validation and Sanitization.
  -> Prevent attackers from injecting malicious data into your system by ensuring that all inputs are validated and sanitized.
  -> ~~X~~SS (cross-site scripting) :- Attackers can inject malicious scripts into web pages. Encode output to prevent xss.
- Authentication and Authorization
  -> Use strong password policies
  -> Implement multi-factor authentication (MFA).
  -> Use RBAC (Role-based access control).
  -> Never dispose senitive data in tokens.
- ~~Data~~ CSRF (Cross-site Request Forgery)
  -> Use anti CSRF tokens.
  -> Ensure that tokens are unique per session and validated on the server side.
  -> Apply the same-site cookie attributes to cookies to prevent CSRF.
- Logging and Monitoring
  -> Use centralized logging system.
  -> Enable application logging at appropriate levels (info, warning, error)
  -> Monitor logs for anamolies or signs of attacks.
- Data Protection
  -> Implement data backup and recovery.
  -> Validate data integrity.
  -> Use secure protocols for data transfer.

Scanned with OKEN Scanner

- **Network Security**
  - → Implement firewall rules
  - → Configure secure DNS
  - → Monitor network traffic
  - → Regularly update network devices.

- **Testing and Deployment**
  - → Perform regular security audits
  - → Conduct penetration testing
  - → Monitor application performance.
  - → Implement incident response plans.

(2) Create a Detailed comparison b/w Web Browser and web server.

⇒

Aspects.	Web browser	Web server.
Role.	Client that requests, retrieves, renders. web content	Server that precess. requests and sends back content.
Primary function	Display and interact with web content	Server web pages & resources.
Data Processing	Processes client-side code (HTML, CSS, JS).	Processes server-side code and logic.
Request/Response	Sends HTTP requests, receives and renders response.	Receives HTTP request sends back responses.
Interaction	Direct interaction with users through a graphical interface	Indirectly interacts by responding to browser requests.
Caching	Caches frequently used content for faster access.	Caches responses for faster delivery to clients.
Applications.	Chrome, Edge, FireFox	Apache HTTP server, Nginx, Tomcat, nodejs.
Development process.	Easy.	Complex.

Scanned with OKEN Scanner

(5) Write JS program and explain advantages and Disadvantages along with it's features.

Page No.
Date

⇒
```
<!DOCTYPE html>
<html>
    <head>
        <title> Simple form Validation </title>
        <script type = "text/javascript">
            function validateForm()
            {
                var email = document.getElementById("email").
                                                value ;
                var password = document.getElementById
                                ("password").value;
                if (!email.include('@') || !email.includes('.'))
                {
                    alert("Please enter valid email.");
                    return false;
                }
                if (password.length < 8).
                {
                    alert("password contain > 8 char long");
                    return false;
                }
                return true;
            }
        </script>
    </head>
</html>
<body>
    <form onsubmit = "return validateForm()">
        <label for = "email"> Email : </label>
        <input type = "email" id = "email" name = "email">
        <label For = "password"> Password </label> <br>
        <input type = "password" id = "password"
                name = "password"> <br> <br>
        <input type = "submit" value = "submit">
    </form>
</body>
</html>
```

Scanned with OKEN Scanner

Advantages

Disadvantages

Page No.
Date

→ Advantages

* Interactivity → Enhances web pages by enabling dynamic and interactive content such as animations, form validation & event handling.
* Cross-Browser compatibility → Modern JS frameworks and libraries help make code compatible across different browsers.
* Versatility → JS can be used for both front-end & back end development, which allows full stack development.

→ Disadvantages

* Security Risks → JS runs on the client side, it can be exploited by malicious actors to inject harmful scripts. (XSS).
* Browser Dependence → JS behaviour may vary slightly b/w browsers, leading to compatibility issues despite modern standards.
* Single-Threaded → JS is single threaded, it handles tasks one at a time.

→ Features

* Interpreted Language
* Client-side scripting
* Dynamic Typing
* Prototypal Inheritance
* Event-Driven Programming
* Asynchronous Programming
* Support for DOM Manipulation.
* Support For JSON
* Functional programming.

Scanned with OKEN Scanner

(11) Explain Events & Event handling in JS with ex.

=> Events and event handling in JS are fundamental concepts that allow developers to create interactive web applications.

An event is any action or occurrence that can be detected by the browser, such as mouse click, a key press, or a page load.

Event handling refers to the process of responding to these events by executing specific code when they occur.

Ex:-

```
<!DOCTYPE html>
<html>
<head>
    <title> Mouseover & Mouse out. </title>
</head>
<body>
    <img id="image" src="boy.jpg"
    onmouseover = "changeImage()"
    onmouseout = "revertImage()">

    <script>
        function changeImage()
        {
            document.getElementById('image').src='girl.jpg';
        }
        function revertImage()
        {
            document.getElementById('image').src='boy.jpg';
        }
    </script>
</body>
</html>.
```